Anugyan Sharma

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Bioengineering undergraduate at the UC San Diego with research experience in biomechanics, biomedical sensing, and clinical data science. Skilled in 3D human-motion analysis, computational modeling, and hands-on engineering, supported by coursework in biomechanics, biomaterial/bioinstrumentation design, mechanical design, Python, MATLAB, statistics, circuits, and advanced math.
Research Interests: Biomechanics, biomedical instrumentation, wearable physiological monitoring systems, bio-sensing technologies, biomedical signal processing, and medical device development for clinical applications.

EDUCATION

University of California, San Diego — **La Jolla, CA**
B.S. Biomedical/Bioengineering — Graduation: June 2027

- Coursework: Musculoskeletal and Soft Tissue Biomechanics, Biomaterial and Bioinstrumentation Design, Mechanical Design Lab, MATLAB, Python, Probability and Statistics, Thermodynamics and Kinetics, Circuits, Arduino Lab
- Involvement: Vice President of Recruitment for Phi Delta Epsilon Medical Fraternity, Triton Prosthetics Society, UCSD Biomedical Engineering Society, IEEE, Surf Club, Sangam, UCSD Outback Adventures, Da Real Punjabiz Bhangra Team

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EXPERIENCE + PROJECTS

Triton Prosthetics Society - UC San Diego — **La Jolla, CA**
Electrical Team Member — Feb 2026 - Present

- Designed and implemented the power architecture for a low-cost motorized above-knee prosthetic leg
- Evaluated battery chemistries (LiPo, Li-ion, SLA, tool packs) based on voltage headroom, discharge rate (C-rating), weight, cost, and motor transient requirements.
- Analyzed motor power demands (~24V, 2–3A transient loads) and implemented voltage filtering and decoupling strategies to mitigate stepper-induced electrical noise
- Contributed to closed-loop control system integrating IMU and load cell feedback for gait phase detection and PID-controlled knee actuation

UC San Diego Men's Basketball × Wu Tsai Human Performance Alliance — **La Jolla, CA**
Performance Analytics Intern — Dec 2025 - Present

- Built an automated data pipeline converting Genius Sports XML feeds into structured Smartabase datasets to support athlete workload monitoring and injury risk analysis.
- Applied data validation, normalization, and schema alignment to ensure accuracy and consistency across performance datasets.
- Enabled scalable ingestion of competition data to support decision-making in athlete readiness, workload management, and injury risk monitoring.
- Collaborated with performance scientists and coaches to ensure data outputs matched operational and reporting requirements.

UC San Diego Health/Department of Psychiatry — **La Jolla, CA**
Clinical AI Research Assistant — Nov 2025 - Present

- Building SQL pipelines to extract and join EMR data (PHQ-9, treatments, diagnoses, demographics) from the UC Health Data Warehouse (>10M patients).
- Preparing cleaned, analysis-ready datasets to model patient response to rTMS vs intranasal Esketamine for treatment-resistant depression.
- Supporting development of machine-learning models predicting treatment response and symptom trajectories.
- Performing time-to-response and survival analyses using repeated PHQ-9 assessments across multi-site clinical populations.

SKILLS

- Technical: Python, MATLAB, SQL, R, Java, CAD, ImageJ, Power BI, Minitab, Arduino, GPU Computing, WSL/Linux, ZMQ, Git
- ML & Data: 3D Pose Estimation, Computer Vision, Random Forest, SVM, Logistic Regression, Survival & Time-Series Analysis, EMR Data Processing, PHQ-9 Modeling
- Lab: Microscopy, Pipetting, Gel Electrophoresis, Spectrophotometry, Fluorescence Assays, Cell Culture, Sterile Technique, Tissue Dissection, Motors & Circuits, Prototype Fabrication, Physiological Sensors (ECG, GSR)